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Draft translation - not yet human-reviewed / مسوده ترجمه

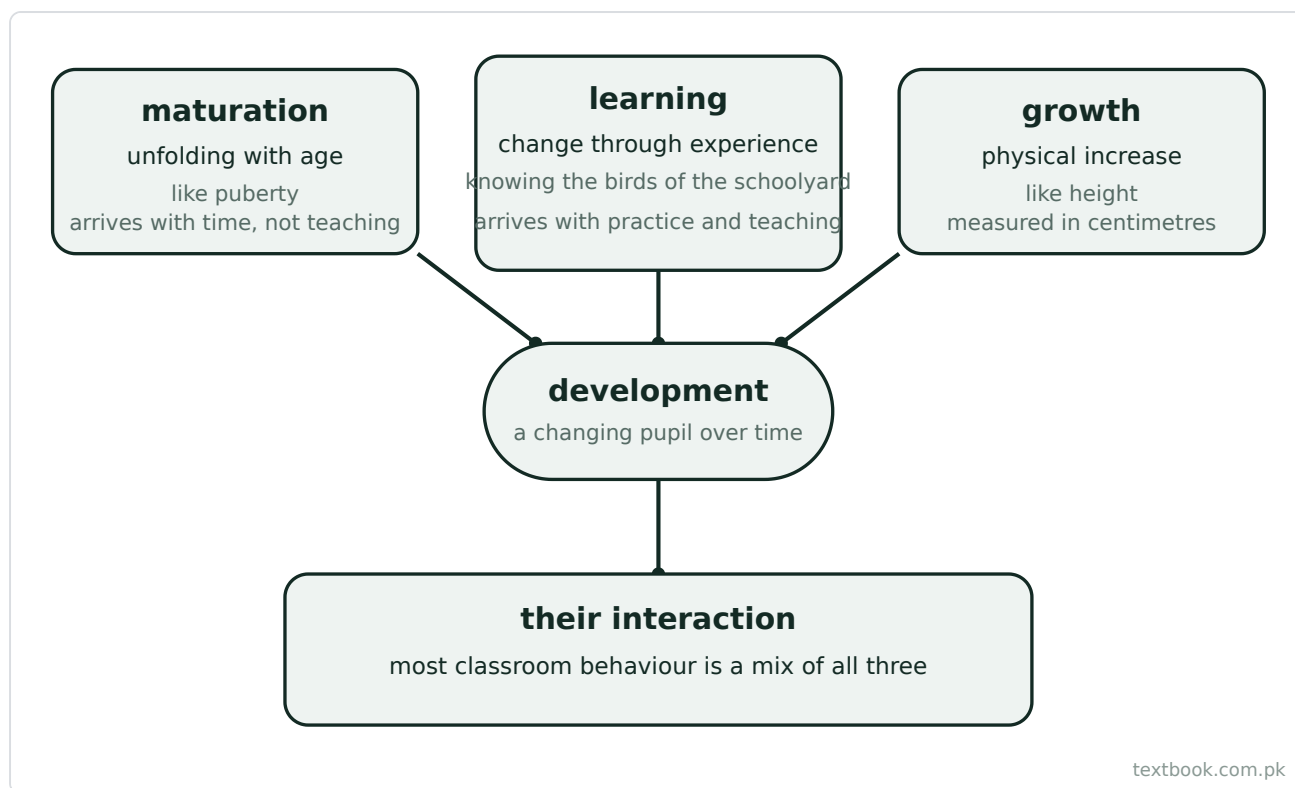
Topic 2.1 - What development is, and the principles it follows

A real classroom situation

Miss Rubina teaches Class 3 in a government primary school on the edge of Hyderabad. In March she received a new pupil, Allah Dino, who had moved from a village school in Tando Muhammad Khan. He could copy any sentence she wrote on the board, letter by letter, in a careful hand. He could not compose a sentence of his own, and when she asked him to read back what he had copied, he read the letters, not the words.

The staff room had its explanations ready by the end of the week. "He is weak." "Village schools teach nothing." "His family never sent him properly." Miss Rubina noticed something none of the explanations covered. Allah Dino tied his shoelaces faster than anyone in the class. He could name every bird in the schoolyard and say which ones came in which season. And his copying was improving, week by week, without anyone teaching him writing directly.

Whatever was going on, it was not one thing. Some of it he had never been taught, some of it was arriving with time, and some of it - the birds, the laces - had clearly been learned somewhere else entirely.



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Explanation

What development is: growth, maturation and learning

Development is the word psychologists use for long-term, orderly change in a person - change in body, thinking, feeling and relationships as age advances (Seifert & Sutton, 2009). It is a broad word, and it hides three different engines that teachers constantly confuse.

Growth is physical increase: getting taller, heavier, stronger. It is measured in centimetres and kilograms, and it is the engine a teacher has least influence over.

Maturation is the unfolding of a built-in programme with age. Puberty is the clearest example: it arrives on its own schedule, whatever the school does. You cannot teach a child to reach puberty, and you cannot hold it back.

Learning is change through experience: what practice, teaching and encounters with the world add. Allah Dino's knowledge of birds is learning. So is

his copying, which is improving because he practises it daily.

The reason to keep the three apart is that each calls for a different teacher response. A pupil who cannot do something because of maturation needs **waiting**, not remedial drills. A pupil who cannot do something because of missing learning needs **teaching**. A pupil whose difficulty is neither - who is simply smaller, because of growth - may need nothing at all except a desk that fits. Allah Dino's sentence-writing was almost certainly a learning gap from his previous school; his careful hand showed the maturation was in place; his reading of letters instead of words was a second, separate learning gap. One label - "weak" - was covering three different situations.

The misconception to clear now. Many trainees arrive believing development means "growing bigger". It does include that, but a pupil who is taller is not thereby more developed in any sense a teacher cares about. The word covers the whole long-term package: body, thinking, feeling and relationships together. When this course says a pupil "is developing", it never means only that the pupil is getting taller.

The principles: continuous yet stage-like, orderly in sequence, uneven in rate

Development follows a small number of principles so reliable that teachers can plan around them (Seifert & Sutton, 2009).

It is continuous, yet it arrives in stages. Change happens every day, but it is not a smooth slope. Long plateaus are punctuated by bursts. A Class 4 pupil may seem to make no progress in reading for four months and then jump two reading levels in three weeks. The jump was prepared the whole time; it just became visible at once. The "staircase" image - flat steps, then a rise - describes this better than a ramp.

The sequence is orderly. The order of development is essentially the same for everyone. Children babble before they speak words, speak words before sentences, draw a person as a head with legs before a head with a body. No child skips a step, whatever the staff room claims about prodigies. What differs between children is not the order but the **speed**.

The rate is uneven - between pupils, and within one pupil. Two children with the same birthday can be a full year apart in reading, and the same child can be ahead in language and behind in number. This is the principle a teacher breaks most often, because schools group pupils by age and then quietly expect them to develop at the same speed. Allah Dino's classmates were all nine; that told Miss Rubina almost nothing about where each of them was.

Try this on the evidence. Ask a Class 1 and a Class 5 pupil in any school to draw a person. The Class 1 drawing is usually a head with legs sticking out of it. The Class 5 drawing has a body, arms, clothes, perhaps fingers. Every child passes through the first kind of drawing to reach the second; no child draws the detailed figure first and simplifies later. That is sequence. But in any Class 1 you will find one child already drawing bodies and one still drawing head-with-legs. That is rate.

engine	what it is	an everyday example	what a teacher can do
① growth	physical increase, measured in centimetres	a pupil a head taller than last year	adjust the desk and the seating plan
② maturation	a built-in programme unfolding with age	puberty arriving on its own schedule	wait, and prepare the pupil for it
③ learning	change through experience and practice	knowing the birds of the schoolyard	teach it, and check the teaching worked
Three named engines, three different classroom responses			

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Heredity and environment as joint drivers

What makes one pupil taller, another quicker with numbers, a third slower to read? The oldest wrong answer is to pick a side: it is all in the genes, or it is all in the upbringing. Developmental psychology rejected that contest long ago.

Heredity supplies the range of possibilities: the height a body can reach, the temperament a child arrives with, the raw speed of certain mental processes.

The environment decides where inside that range a pupil actually lands: food, health, sleep, talk at home, books, teaching, safety. Neither works alone. A pupil inherits a potential height, but chronic malnutrition keeps the body well below it. A pupil inherits no vocabulary at all - every word, in every language, is learned from the environment.

The classroom version of the principle matters more than the biology. Two sisters attend Miss Rubina's school. The elder is tall for her age; the younger is not. Height ran in the family, but so did evening story-telling, and the younger sister, who shares a room with the elder and hears the same stories, has the larger vocabulary. Which is heredity and which is environment? The honest answer is that the question is badly formed. Both contributed to both outcomes, and no observation in a classroom can separate them.

What a teacher takes from this is a posture, not a formula: **every pupil in front of you is a joint product, and you control part of the environment.** You cannot change a pupil's heredity, and you cannot change most of the environment either. You can change what happens in your room for six hours a day, which is not nothing. The pupils who most need that six hours to be good are, reliably, the ones the staff room has labelled with the heredity explanation: "that family is just like that."

Activity: Sort the changes

In pairs, about 20 minutes.

Below are ten changes Miss Rubina might observe in a pupil over a school year. Decide for each whether it is best explained by **growth, maturation, learning**, or a mix. For each, say what Miss Rubina should do about it, if anything.

1. The pupil can suddenly reach the top shelf of the class library.
2. The pupil's voice breaks in the middle of a recitation.
3. The pupil can now subtract with borrowing, after three weeks of practice sums.
4. The pupil's drawings of people gain fingers and shoes.

5. The pupil stays awake through the afternoon lesson for the first time.
6. The pupil starts raising a hand before speaking, after a class rule was introduced.
7. The pupil grows out of the uniform by March.
8. The pupil can catch a cricket ball struck hard and low.
9. The pupil uses "could" instead of "can" when speaking to the head teacher.
10. The pupil stops crying at wrong answers.

Then pick the **one** change where you and your partner disagreed about the engine, and write two sentences: one making the strongest case for your reading, one for your partner's.

Check your understanding

1. In your own words, define development in one sentence, without using the words "growth", "bigger" or "change over time". (*Understand*)
2. Name the three engines this topic distinguishes, and give one classroom example of each from a pupil you have actually taught or been. (*Understand*)
3. A colleague says "Allah Dino is weak." Using the three engines, give two different explanations of the same observation that would lead to different next steps. (*Apply*)
4. Explain the difference between sequence and rate, using the person-drawing evidence this topic describes. (*Understand*)
5. A parent tells you "his whole family was slow at reading, so he will be too." Give the fairest reply you can, in three sentences, using the heredity-and-environment principle. (*Apply*)

Summary

Development is long-term, orderly change across body, thinking, feeling and relationships, driven jointly by growth, maturation and learning. Its sequence is essentially fixed - every child passes the same steps in the same order - but its rate is individual, so an age-grouped class contains a wide spread. Heredity sets ranges and environment fills them, and no classroom observation can divide the

two. For a teacher, the practical shift is from "why is this pupil not like the others?" to "where is this pupil, and what is the next step from there?"

Self-assessment checklist

Sign in to sync your ticks across devices.

- ☐ I can define development and name its three engines with a classroom example of each.
- ☐ I can explain what the staircase image captures that a ramp does not.
- ☐ I can state the difference between sequence and rate and give evidence of each.
- ☐ I can explain why "heredity or environment?" is a badly formed question, and what a teacher controls regardless.

Try this at your practicum school

Ask the cooperating teacher for one piece of work from a Class 1 pupil and one from a Class 5 pupil on the same kind of task (a drawing of a person, or a short piece of writing). Compare them against the principles of this topic: what is the same in the sequence, and what differs in the rate? Note one thing the older pupil can do that the younger one will do later, without being taught exactly how.

Summative task

Write **250-300 words** on Allah Dino.

Take the three staff-room explanations ("weak", "village schools teach nothing", "his family never sent him properly") and the reading Miss Rubina reached. For each explanation, say which engine of development it assumes, what evidence from the topic's description supports or counts against it, and what it tells Miss Rubina to do next. Conclude with the explanation you would act on first and why.

Mini-rubric

Criterion	What a strong answer shows
Use of the three engines	Each explanation is read against growth, maturation and learning rather than accepted or rejected wholesale.
Evidence discipline	Claims about Allah Dino cite the described observations (laces, birds, improving copying), not imagined ones. <i>(Analyze)</i>
Teaching judgement	The chosen next step follows from the analysis and could be done by one teacher in one classroom.
Plain, accurate writing	Terms used correctly; no padding; within the word count.

Further reading

- Seifert, K., & Sutton, R. (2009). *Educational Psychology* (2nd ed.), Chapter 3. University of Manitoba. CC BY. <https://open.umn.edu/opentextbooks/textbooks/educational-psychology>
- Vosniadou, S. (2001). *How children learn*. Educational Practices Series No. 7. International Academy of Education / UNESCO IBE. <https://eric.ed.gov/?id=ED455955>

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